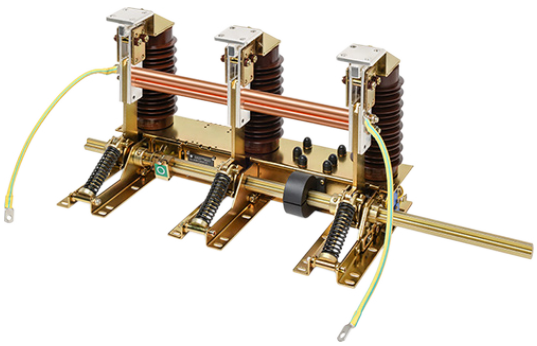
● GPJN15-12(24)/D31.5
Indoor high voltage motorized earthing switch
GPJN15-12/D40, 50
Indoor high voltage motorized earthing switch
GPJN15-40.5/D31.5
Indoor high voltage motorized earthing switch



GPJN15-12(24)(40.5)kV motorized earthing switch

GPJN15-12(24)/D31.5, GPJN15-12/D40, 50, GPJN15-40.5/D31.5 series indoor high voltage motorized earthing switch (short as: motorized earthing switch) is designed for the intelligent switchgear requirement of the electric system. It is suit for 3~40.5KV, 3 phase, AC 50 (60) Hz power system and kinds of high voltage switchgears. It is applicable for high voltage switchgear arrangement as the earthing protection when maintenance.
The products' performance meet GB1985-2004 High Voltage AC Disconnector and Earthing Switch and the requirements of standard as well as IEC62271-102:2002.

Earthing switch structure is simple and compact, light weight, flexible operation, easy installation, dynamic thermal stability, and a reliable electric operation, keep the original interlocks in catching rabbits switchgear chain constant, has been a national patent.

Products summary

Due to the earthing switch is electrically operated, The cabinet manually operated interlocks is still keep the same as original design, but there are some parts have been modified, be sure to use our company's matchable connection rod , otherwise it would be burned the motor of motorized earthing switch.
Since the earthing switch is electrically operated, and therefore the cabinet manually operated chain catching rabbits consistent with the original style, but there are some parts have been modified, be sure to use the matching connection rod of our company, otherwise it would be burned the motor of motorized earthing switch.
If the cabinet back door need interlocking with motorized earthing switch, also be sure to use the company's patented technology to produce backdoor interlock mechanism, and equipped travel switch.
When the back door is not closed, cutting off the operation power of motorized earthing switch to avoid motorized earthing switch of misuse and damage the motor.
When electric interlocking with the motorized earthing switch, be sure to use auxiliary contacts associated with electrical components to disconnect the operation power of the motorized earthing switch to achieve electrical interlock in order to ensure a reliable electrical interlock.

Products model and its meaning

GP J N 15 - 24 / D 31.5 - □

- Phase distance (mm)
- Rated short-time withstand current
- Motorized (electric)
- Rated voltage12kV, 24kV, 40.5kV
- Design No.
- Indoor
- Earthing switch
- Company feature code

Basic parameters

Products model		GPJN15			
Rated working voltage		12kV	24kV	40.5kV	
Rated short-time withstand current (4S)		31.5kA	40kA	50kA	31.5kA
Rated peak withstand current		80kA	100kA	125kA	80kA
Rated short-circuit close current		80kA	100kA	125kA	80kA
Insulating level:	Main circuit withstand voltage(1min)	42kV	65 kV		95kV
	Rated lighting impulse withstand voltage	75kV	125 kV		185kV
Across distance after open not less than		125mm	200 mm		300mm
Control voltage		AC220V	DC220V		DC110V

Standard working conditions

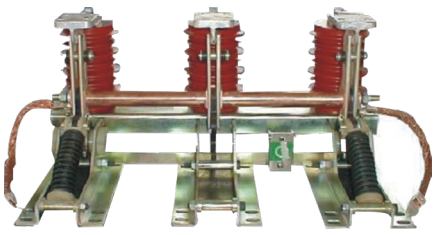
Apply to the following conditions:
1. Not more than 1000 meters above sea level;
2. Ambient temperature not higher than + 45℃ , not lower than -25℃ ;
3. Daily average relative humidity less than 95%, the monthly average of less than 90%;
4. Earthquake intensity does not exceed 8 degrees;

Not available the following locations:
1. Installation site there is a serious grounding switch insulation and conductivity of gases, vapors, chemical precipitation, dust, dirt and smoke explosion aggressive substances;
2. Installation place has frequent violent vibration.

Structure and working principle

1. Main structure
The earthing switch mainly formed by the holder, ground knife assembly, Fixed contact, sensors, shafts, crank arm, spring, conductive sleeve, flexible connection, curved rear bracket by a thin steel plate welded into parts, the rest are assembled components. Its shape, see the following picture:

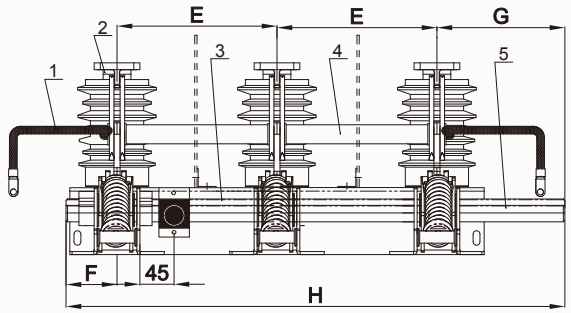
Figure 1: Outline of the motorized earthing switch

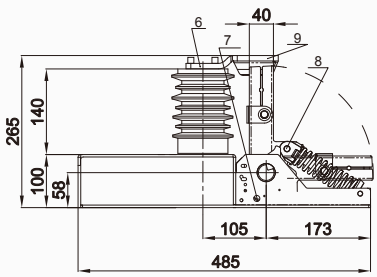


2. Earthing switch works principles (in Figure 2, for example)
When the manually operated earthing switch closed, the torque force to the shaft 5 to overcome the drag torque, driving crank arm 7 rotates in the closing direction, and the operation rod on the earthing knife 9 through pressure spring 8 dead point, spring release energy, the grounding switch rapidly closing, on the closing position, earthing knife of earthing knife components 9 through the butterfly spring and knife mouth of fixed contact 6 to earthing firmly and securely.
When manual operation opening, the torque force to the shaft 5 and the spring force to overcome the resistance torque, drive crank arm 7 along the opening direction of rotation, and earthing knife 8 and pressure spring 9 through the dead point, spring energy storage finished in order to prepare for the next closing times.
When electric operation, earthing switch inside has bevel gear, reducer, motor, control board which combined to drive device, this device output torque forward and reverse directions, so that the earthing switch opening or closing.
Whether earthing switch is manually or electrically operated, it is regardless of the speed of closing and opening the earthing switch.

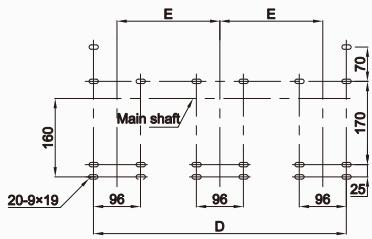
Outline dimension, installation dimension and busbar joint dimension

Figure 2: GPJN15-12/D31.5/150 motorized earthing switch outline dimension





Holing dimension



Products model	E	F	G	H	D
GPJN15-12/D31.5-150	150	75	140	515	396.5
GPJN15-12/D31.5-165	165	75	160	565	426

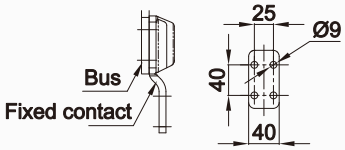
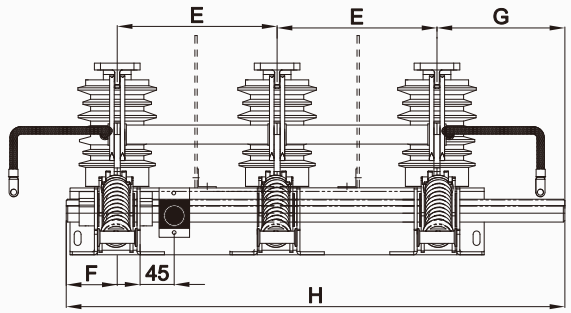
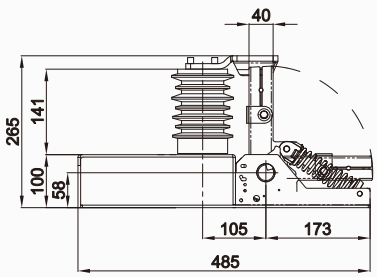
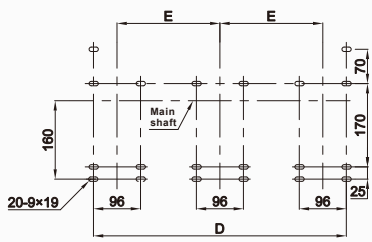


Figure 3: GPJN15-12/D31.5/210/275 motorized earthing switch outline dimension



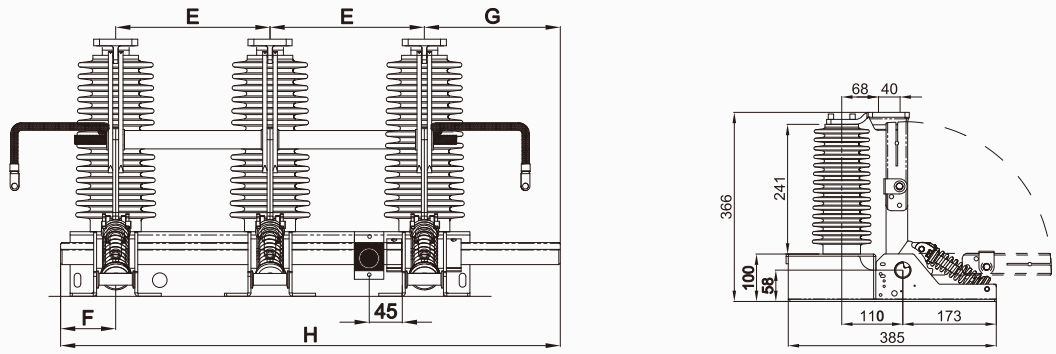


Holing dimension

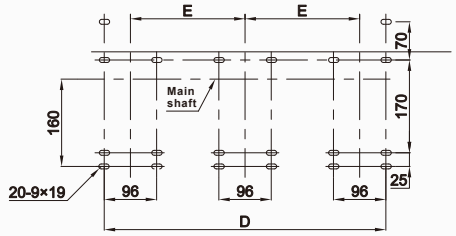


Products model	E	F	G	H	D
GPJN15-12/D31.5-275	275	75	185	810	646
GPJN15-12/D31.5-210	210	75	160	665	516
GPJN15-12/D31.5-250	250	75	160	735	596

Figure 4: GPJN15-24/D31.5/210/275 motorized earthing switch outline dimension



Holing dimension



Products model	E	F	G	H	D
GPJN15-24/D31.5-275	275	75	185	810	646
GPJN15-24/D31.5-210	210	75	160	665	516

Phase distance 210mm need add insulating partition

Figure 5: GPJN15-12(24)/D31.5

GPJN15-40.5/D31.5 series motorized earthing switch busbar joint size

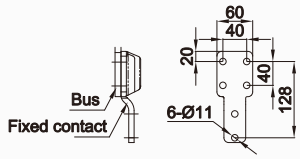
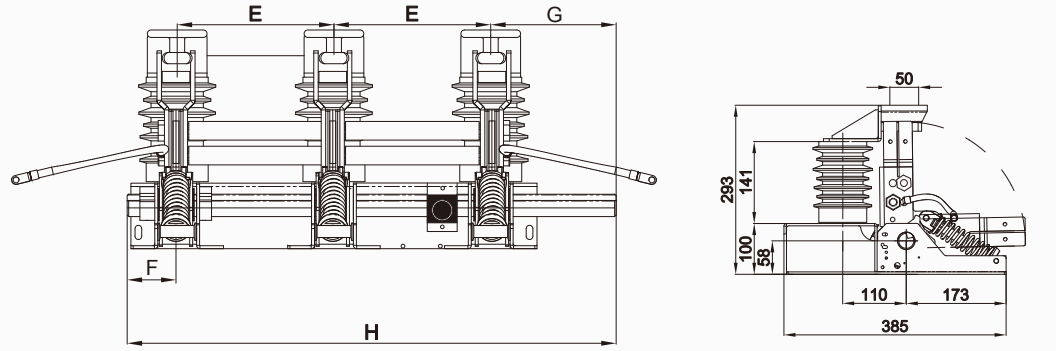
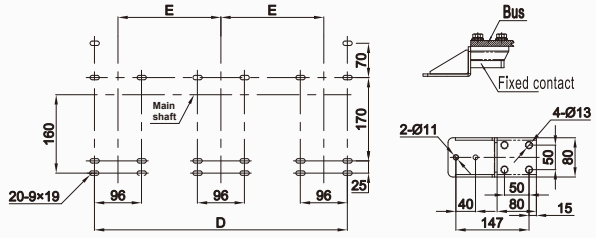


Figure 6: GPJN15-12/D40/210/275 motorized earthing switch outline dimension

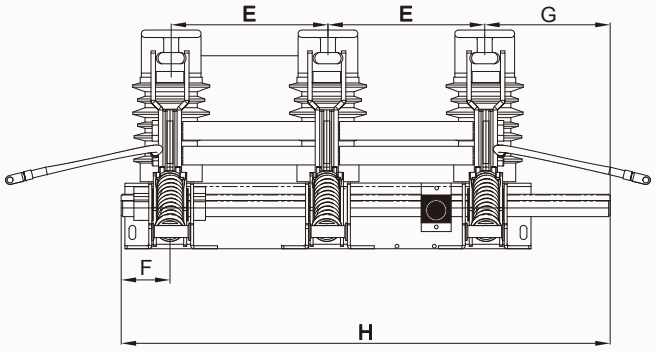


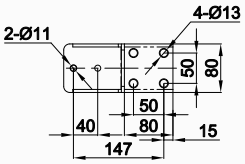
Holing dimension

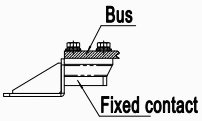


Products model	E	F	G	H	D
GPJN15-12/D40-275	275	75	185	810	646
GPJN15-12/D40-210	210	75	160	665	516

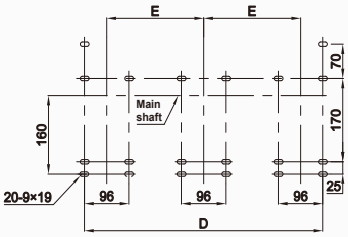
Figure 7: GPJN15-12/D50/210/275 motorized earthing switch





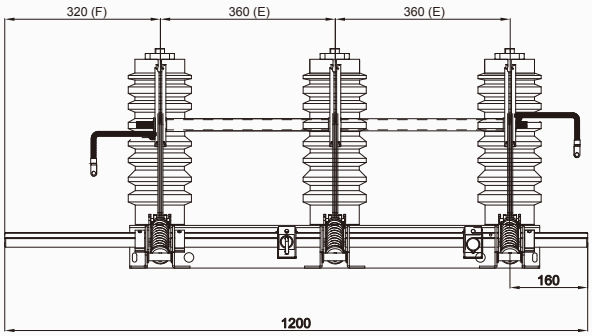


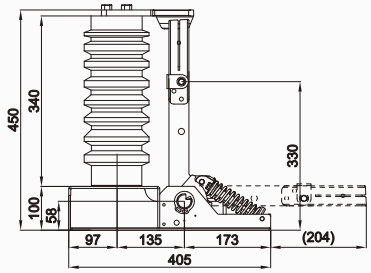
Holing dimension

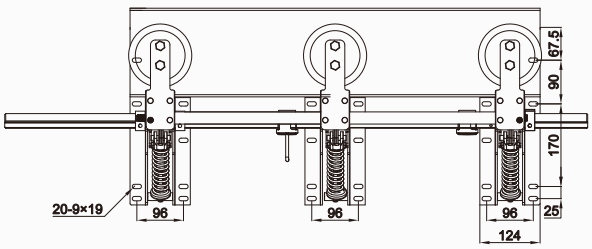


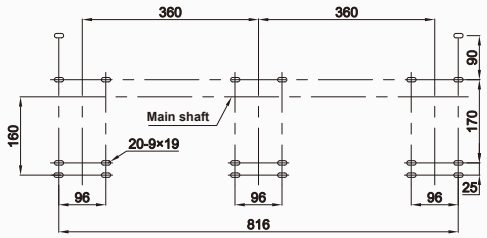
Products model	E	F	G	H	D
GPJN15-12/D50-275	275	75	185	810	646
GPJN15-12/D50-210	210	75	160	665	516

Figure 8: GPJN15-40.5/D31.5 motorized earthing switch outline dimension









Products model	E	F
GPJN15-40.5/D31.5-280	280	290
GPJN15-40.5/D31.5-300	300	340
GPJN15-40.5/D31.5-350	350	340
GPJN15-40.5/D31.5-360	360	320
GPJN15-40.5/D31.5-400	400	340

Note: if phase distance 300 need add insulating plate between phases.

